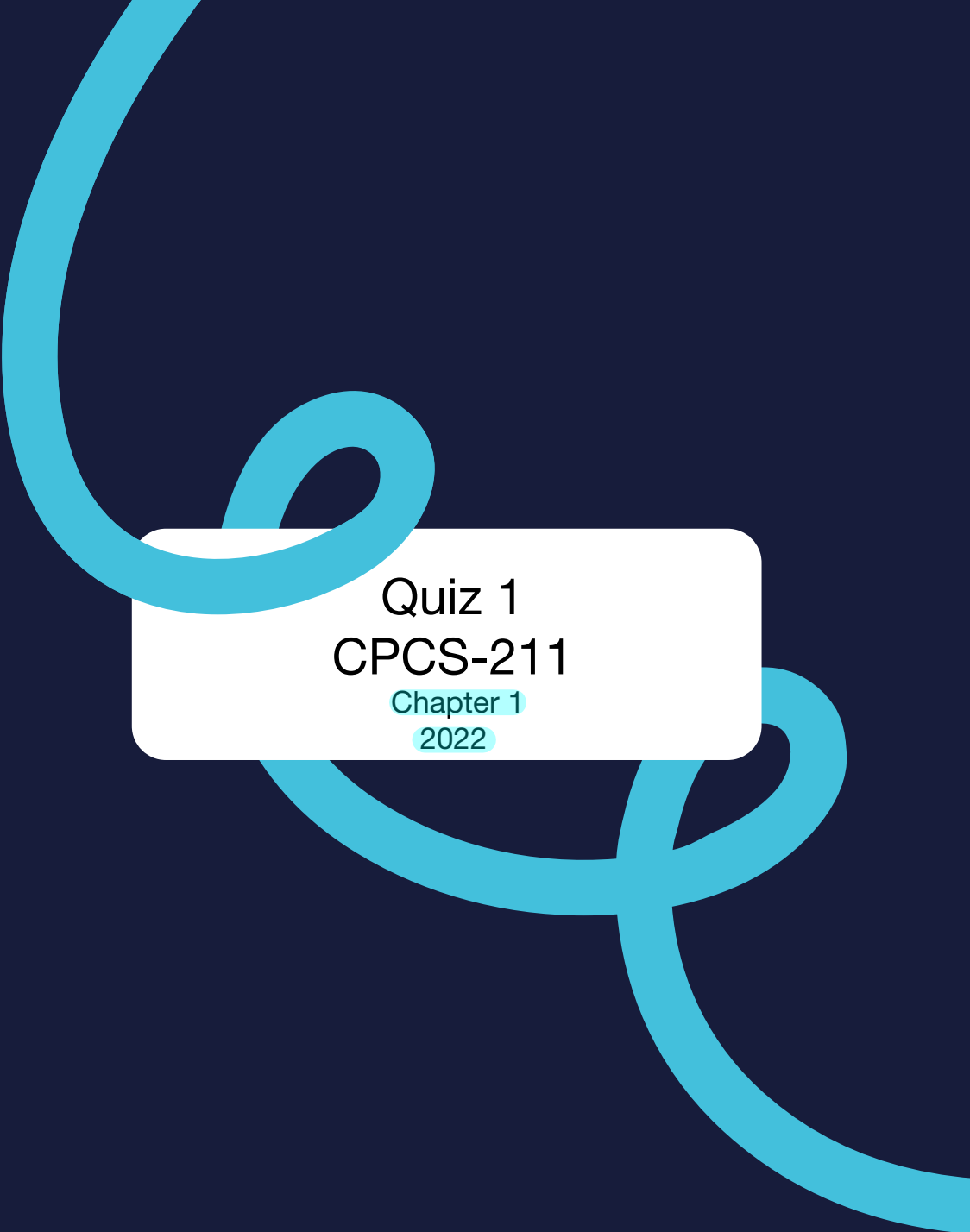




Quiz 1

CPCS-211

Chapter 1
2022

Convert the decimal number 250.82 into corresponding two's complement binary format.

Use 6 digits of precision to the right of the binary point?

- ☐ 1111 1010.010001
- ☐ 1010 1111.0100 10
- ☐ None
- ☒ 1111 1010.0100 10

What is the sum of the following two's complement numbers?

- 11110010
01101110
- ☐ 10011000
 - ☐ 10110000
 - ☐ None
 - ☒ 01100000

A computer with a 16-bit word size uses 2's complement to represent numbers.

The range of integers that can be represented by this computer is

- ☐ -2^{16} to 2^{16}
- ☐ None
- ☐ -2^{16} to $2^{16}-1$
- ☒ -2^{15} to $2^{15}-1$

What the BCD representation of decimal number 125?

- ☒ 0001 0010 0101
- ☐ 0111 1101
- ☐ 125
- ☐ None

What is the octal equivalent of hexadecimal number 05BC ?

- ☐ 1468
- ☐ 1336
- ☐ None
- ☒ 2674

What is the excess 3 representation of decimal number 364 ?

- ☒ 0110 1001 0111
- ☐ 0011 0110 0100
- ☐ 367
- ☐ None

Perform the following two's complement arithmetic operation in the way a computer would perform it.

1110 0110 1100

1100 1001 1011

Indicate if an overflow occurred?

- ☐ None
- ☒ No overflow occurred
- ☒ Yes overflow occurred
- ☐ Borrow occurred

In 2's complement representation, a certain negative number $-N$ is represented

- ☐ None
- ☐ 0000 0110 1011
- ☐ 148
- ☒ 0000 0110 1100